

TraceBridge Medical - Horizon POD System

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1. Introduction and Scope

This Risk Management Plan describes the process for managing the risks associated with the Horizon POD System, a continuous subcutaneous insulin infusion (CSII) pump. The scope of this plan covers the entire lifecycle of the device, from design and development through post-market surveillance and eventual decommissioning.

2. Intended Use and Foreseeable Misuse

Intended Use: The Horizon POD System is intended for the continuous delivery of insulin, at set and variable rates, for the management of diabetes mellitus in persons requiring insulin.

Foreseeable Misuse:

- User dropping the POD from a height greater than 1 meter.
- User exposing the POD to extreme temperatures (e.g., hot tubs, freezing environments).
- User ignoring low reservoir or occlusion alarms.
- User attempting to modify the POD reservoir with non-standard insulin concentrations.

3. Risk Management Process

The risk management process follows a lifecycle approach, incorporating hazard identification, risk estimation, risk evaluation, risk control, and post-production monitoring.

3.1 Hazard Identification

A comprehensive hazard analysis was conducted on April 12, 2026, involving representatives from Engineering, Quality Assurance, and Clinical Affairs. The following broad hazard categories were evaluated:

- **Energy Hazards:** Electrical shock, thermal energy (battery overheating).
- **Biological/Chemical Hazards:** Biocompatibility of skin-contacting materials, insulin degradation.
- **Operational Hazards:** Over-delivery of insulin, under-delivery of insulin, occlusion.
- **Information Hazards:** Incorrect display of remaining insulin, delayed alarms.

3.2 Identified Hazards and Hazardous Situations

The team has documented the following primary hazardous situations:

1. **Hazardous Situation 01:** Software failure leading to continuous motor drive activation, resulting in massive insulin overdose.

2. **Hazardous Situation 02:** Cannula occlusion goes undetected by the pressure sensor, resulting in insulin under-delivery and potential diabetic ketoacidosis (DKA).
3. **Hazardous Situation 03:** Bluetooth communication drop between the POD and the mobile controller during a bolus command, leading to uncertain delivery state.
4. **Hazardous Situation 04:** Lithium-ion battery thermal runaway due to short circuit, causing patient burn.

4. Risk Acceptability Criteria

Note: The cross-functional team is still debating the threshold for ALARP (As Low As Reasonably Practicable). We have temporarily suspended the creation of the Risk Evaluation Matrix (Probability vs. Severity). The QA Director has advised that we can proceed with hazard listing and implement mitigations without formalizing the mathematical probability scores for this submission, as long as we show we addressed the hazards.

5. Risk Control Measures

For the identified hazardous situations, the following controls have been implemented:

- **Control for HS-01 (Overdose):** The motor controller circuit includes a hardware watchdog timer. If the microcontroller hangs while the motor is active, the watchdog will mechanically sever power to the motor driver after 2.5 seconds.
- **Control for HS-02 (Occlusion):** A micro-electro-mechanical systems (MEMS) pressure sensor is installed in the fluid path. The firmware samples this sensor every 500ms. An occlusion is flagged if the pressure exceeds 15 PSI for more than 3 consecutive readings.
- **Control for HS-04 (Thermal):** The battery management system (BMS) includes an integrated thermistor. Charging and discharging are disabled if the cell temperature exceeds 45°C.

6. Post-Production Information

Post-market surveillance will involve the systematic review of customer complaints, MAUDE database queries for similar CSII devices, and clinical literature reviews every 12 months.

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